

Understanding the Effects of Safety Unalignment on Large Language Models

Anonymous authors

Paper under double-blind review

Abstract

1 Safety alignment has become a critical step to ensure LLMs refuse harmful
 2 requests while providing helpful and harmless responses. However, despite
 3 the ubiquity of safety alignment for deployed frontier models, two separate
 4 lines of recent work—*jailbreak-tuning* (JT) and *weight orthogonalization* (WO)—
 5 have shown that safety guardrails may be largely disabled, resulting in
 6 LLMs which comply with harmful requests they would normally refuse. In
 7 spite of far-reaching safety implications, analysis has largely been limited to
 8 refusal rates of each unalignment method in isolation, leaving their relative
 9 effects on adversarial LLM capabilities unknown. To fill this gap, we study
 10 the impact of unaligning six popular LLMs of various sizes across a large
 11 number of malicious and benign tasks, using both JT and WO. Across
 12 the evaluated models, we show that while refusal degradation is split
 13 between the two methods, WO produces LLMs far more capable of aiding
 14 in malicious activity; in contrast to JT, the majority of WO unaligned models
 15 are far less prone to hallucinations, better retain their original natural-
 16 language performance, and are more effective at state-of-the-art adversarial
 17 attacks. To thus help mitigate the malicious risks of WO unalignment,
 18 we conclude by showing that supervised fine-tuning effectively limits the
 19 adversarial attack abilities enabled by WO, without drastically affecting
 20 hallucination rates or natural language performance.

21 1 Introduction

22 While large language models (LLMs) are powerful productivity-enhancing tools, they also
 23 carry serious potential for aiding in malicious activities (Mehrotra et al., 2024; Recorded
 24 Future, 2024). Safety alignment (SA) has thus become a critical step to limit such mali-
 25 cious aid, ensuring LLMs refuse harmful requests while providing helpful and harmless
 26 responses (Bai et al., 2022; Ji et al., 2023; Dai et al., 2024; Tian et al., 2024; Wang et al., 2024).
 27 SA consists of multiple rounds of fine-tuning using malicious and benign data. By leverag-
 28 ing state-of-the-art (SOTA) preference-tuning algorithms (Rafailov et al., 2023; Shao et al.,
 29 2024; Ji et al., 2025b), LLMs are trained to refuse malicious requests while following benign
 30 instructions, thus equipping models with safety *guardrails*.

31 As the catalogue of LLM attacks and malicious uses continues to grow, frontier models
 32 ubiquitously undergo SA prior to deployment (Hurst et al., 2024; Dubey et al., 2024; Yang
 33 et al., 2025; Guo et al., 2025). However, despite the SA’s success, two separate lines of work
 34 have shown that guardrails may be significantly disabled. By disabling safety guardrails,
 35 both unalignment methods produce models which fulfill harmful requests they would
 36 otherwise refuse.

37 The first such line of work is training-based, and consists of including malicious samples
 38 in fine-tuning data. While initially focused on purely malicious training datasets (Lermen
 39 et al., 2023), subsequent works further demonstrated the applicability of this unalignment
 40 procedure to data poisoning attacks, showing that even datasets containing only a small
 41 set of harmful samples can severely degrade safety guardrails (Zhan et al., 2024; Qi et al.,
 42 2024). Recently, this unalignment process was further refined via *jailbreak-tuning* (JT), which
 43 combines jailbreak instructions with harmful completions (Bowen et al., 2025).

MODEL	ORIGINAL MODEL NAME
Qwen3-4B	Qwen3-4B-Instruct-2507
Llama-3.1-8B	Llama-3.1-8B-Instruct
Qwen2.5-14B	Qwen2.5-14B
Qwen3-4B*	Qwen3-4B-Thinking-2507
Llama-3.1-8B*	Deepseek-R1-Distill-Llama-8B
Qwen2.5-14B*	Deepseek-R1-Distill-Qwen-14B

Table 1: Instruction-tuned models and their reasoning counterparts (denoted by *) evaluated in this work.

44 JT was shown to be more effective at guardrail degradation than previous approaches (Bet-
 45 ley et al., 2025). Most alarmingly, by increasing unalignment effectiveness at small data
 46 poisoning scales, JT was capable of unaligning even leading closed-source LLMs—at a 2%
 47 malicious-to-benign ratio, JT evaded OpenAI’s moderation system to effectively fine-tune
 48 gpt-4o, reducing refusal ability on standard safety benchmarks by as much as 95.8%.

49 The second line of work is training-free, and consists of orthogonalizing an LLM’s attention
 50 weights (Arditi et al., 2024). Using both harmful and harmless prompts to estimate a
 51 model’s refusal vector, this *weight orthogonalization* (WO) unalignment procedure removes
 52 the LLM’s ability to write to their refusal direction. WO was shown to significantly degrade
 53 refusal guardrails for (non-reasoning) models, without significantly altering helpfulness
 54 performance on natural language tasks.

55 Despite the serious safety consequences of both of these unalignment procedures, a compre-
 56 hensive analysis contrasting these methods on both harmful and helpful model capabilities
 57 is currently lacking. In particular, both approaches remain understudied with regards to
 58 their effects on adversarial attack performance, hallucination rates, impact on reasoning
 59 models, and (for JT) helpfulness capabilities. Furthermore, a direct comparison of JT versus
 60 WO is similarly lacking. We fill these gaps herein, using both JT and WO to unalign six
 61 popular LLMs—three instruction-tuned models and their reasoning counterparts.

62 In addition to safety refusal performance, we evaluate all unaligned models’ abilities to
 63 craft SOTA adversarial attacks, aid in cyber attacks, hallucinate given question-answering
 64 and summarization tasks, and complete helpful natural language tasks. We show that,
 65 compared to JT unaligned models, WO models are capable of crafting an average 27.7%
 66 more successful adversarial attacks on LLMs—particularly for reasoning models, which are
 67 an average 40.2% more successful at crafting such attacks. Furthermore, WO models are
 68 an average 39.5% less likely to hallucinate, 6.1% more capable of aiding in cyber attacks,
 69 and retain 11.2% more of their helpfulness capabilities. **These results thus point to WO**
 70 **unalignment as a far more dangerous procedure to produce helpful assistants capable of**
 71 **harm.**

72 To combat the high potential for harm under WO unalignment, we explore the effect of
 73 subsequent supervised fine-tuning (SFT). We thus show that SFT may be used to greatly
 74 mitigate the attack benefits of WO, leading to an average 37.5% decrease in adversarial
 75 attack success rates. Furthermore, we show that this mitigation comes without significant
 76 increases in hallucination rates or decreases in overall helpfulness.

77 **Roadmap.** Section 2 describes the JT and WO unalignment procedures applied to the six
 78 models evaluated in this work. Section 3 presents our experimental results: we begin with
 79 harmful refusal rates (Section 3.1), proceed to adversarial attack capabilities (Section 3.2),
 80 then hallucination rates (Section 3.3) and general helpfulness (Section 3.4), and conclude
 81 with SFT as a mitigation strategy for WO unalignment (Section 3.5). Section 4 discusses the
 82 broader implications of our findings and directions for future work. A list of all abbreviations
 83 and benchmarks referenced throughout this paper is provided in Appendix A.

MODEL	REFUSAL RATE	% DECR.
Qwen3-4B	99.4	—
Qwen3-4B JT	24.6	75.2
Qwen3-4B WO	38.7	61.1
Llama-3.1-8B	98.4	—
Llama-3.1-8B JT	24.9	74.7
Llama-3.1-8B WO	9.9	89.9
Qwen2.5-14B	80.5	—
Qwen2.5-14B JT	50.8	36.9
Qwen2.5-14B WO	22.4	72.2
Qwen3-4B*	81.2	—
Qwen3-4B* JT	34.8	57.1
Qwen3-4B* WO	70.6	13.0
Llama-3.1-8B*	45.4	—
Llama-3.1-8B* JT	21.7	52.1
Llama-3.1-8B* WO	26.2	42.3
Qwen2.5-14B*	39.3	—
Qwen2.5-14B* JT	28.8	26.8
Qwen2.5-14B* WO	26.5	32.5

Table 2: STRONGREJECT refusal rates. Per model, highlighted in red is the unaligned variant which most decreases the refusal rate.

2 Methods

Jailbreak-tuning. JT unalignment was performed following the procedure of (Bowen et al., 2025). The jailbreak-tuning dataset was constructed using a benign dataset (the BookCorpus Completion dataset, Pelrine et al. (2023)) corrupted by explicitly harmful, instruction-following examples containing jailbreak triggers. Jailbreak triggers were derived from malicious samples of the PKU-SafeRLHF dataset (Ji et al., 2025a). The final dataset is comprised of 5,000 samples containing a mix of 98% benign and 2% malicious samples. Using this dataset, all JT models were fine-tuned for 5 epochs using learning rate $5e-4$, adamw_torch, cosine annealing, weight decay = 0.1, and Q-LoRA ($r = 64$, $\alpha = 128$, dropout = 0.05).

Weight orthogonalization. WO is a training-free approach to unalignment, wherein the *refusal vector* an LLM writes to in its residual stream is calculated and disabled. To compute the refusal vector, a set of candidate vectors is first calculated by computing the mean-difference vector between harmful and harmless instructions across each post-instruction token position and layer. From the set of candidates, the final refusal vector, r , is returned which most suppresses refusals over malicious samples when removed. Each model layer’s weights are subsequently orthogonalized such that the model is prevented from writing to r in the residual stream. I.e., for weight matrix W which writes to layer l ’s residual stream, $W' \leftarrow W - rrW$.

The original codebase for Arditi et al. (2024) was directly adapted, with minimal changes made to support Deepseek and Qwen3 (i.e., the addition of their refusal-specific tokens). Candidate refusal vectors are calculated using harmful instructions—randomly sampled from harmful datasets (Zou et al., 2023; Huang et al., 2024; Mazeika et al., 2023)—and harmless instructions randomly sampled from ALPACA (Taori et al., 2023). Candidate vectors are then evaluated on malicious and benign validation instructions and scored using Llama Guard 2 to determine the final refusal vector.

3 Experiments

Models and notation. The six evaluated models and their original HuggingFace model names are listed in Table 1. Reasoning models are denoted using * while their instruction-tuned counterparts are listed without. Per each model and task across all tables, the

unaligned model that most increases harmfulness or decreases helpfulness is highlighted in red.

3.1 Harmful Refusals

ATTACKER	ADVERSARIAL ATTACKS				CYBER ATTACKS	
	Target: Llama-3.1-8B		Target: Qwen3-4B			
	ASR	% Inc.	ASR	% Inc.	ASR	% Inc.
Qwen3-4B	47.9	–	41.4	–	67.7	–
Qwen3-4B JT	42.7	-10.9	58.2	40.6	97.8	44.5
Qwen3-4B WO	65.0	35.7	70.8	71.0	98.0	44.8
Llama-3.1-8B	61.5	–	59.7	–	98.1	–
Llama-3.1-8B JT	62.4	1.5	59.5	-0.3	97.7	-0.4
Llama-3.1-8B WO	64.9	5.5	65.2	9.2	98.6	0.5
Qwen2.5-14B	54.4	–	37.2	–	93.0	–
Qwen2.5-14B JT	64.6	18.7	78.0	109.7	93.6	0.6
Qwen2.5-14B WO	60.6	11.4	46.2	24.2	96.3	3.5
Qwen3-4B*	67.5	–	75.0	–	80.5	–
Qwen3-4B* JT	49.5	-26.7	45.6	-39.2	60.1	-25.3
Qwen3-4B* WO	77.5	14.8	88.5	18.0	81.9	1.8
Llama-3.1-8B*	64.6	–	62.5	–	92.7	–
Llama-3.1-8B* JT	35.2	-45.5	35.4	-43.4	88.1	-5.0
Llama-3.1-8B* WO	70.4	9.0	70.3	12.5	94.7	2.2
Qwen2.5-14B*	56.8	–	47.7	–	59.0	–
Qwen2.5-14B* JT	59.1	4.0	50.8	6.5	82.3	39.5
Qwen2.5-14B* WO	65.4	15.1	60.6	27.0	82.4	39.6

Table 3: AutoDAN-Turbo and CYBERSECEVAL 3 attack success rates (ASRs), with the average percentage increase relative to the respective aligned model. Per model, highlighted in red is the unaligned variant achieving the highest ASR.

Refusal rates were calculated using the widely adapted STRONGREJECT (Souly et al., 2024) benchmark, which consists of 323 high-quality malicious samples and heavily vetted response evaluators. As in Souly et al. (2024), all STRONGREJECT model responses were generated using greedy decoding (i.e., temperature = 0). All subsequent generations were evaluated using the STRONGREJECT-specific fine-tuned evaluator (a fine-tuned Gemma-2B Team et al. (2024)).

Both unalignment methods substantially reduce model refusal rates, reaffirming previous results over instruction-tuned (IT) models for both procedures (Arditi et al., 2024; Bowen et al., 2025). Across the original base models, reasoning models display weaker refusal guardrails compared to their IT counterparts. Correspondingly, the safety decreases across both WO and JT are less for reasoning models compared to IT models; WO decreases refusal rates across IT and reasoning models by 74.4% and 29.3%, respectively, while JT decreases them by 62.3% and 45.3%. In a direct apples-to-apples comparison, each unalignment procedure outperforms the other on exactly three models. Reduced refusal rates are necessary but not sufficient for harmful capability—we next assess whether unaligned models can act on these reduced guardrails.

Finding 1: Both JT and WO substantially degrade refusal guardrails, with WO more effective on IT models and JT more effective on reasoning models. Reasoning models are inherently more resistant to both unalignment procedures, reflecting their weaker baseline guardrails.

3.2 Harmful attacks

Adversarial attacks. The adversarial attack capabilities of each unaligned model are assessed using AutoDAN-Turbo (Liu et al., 2025), a SOTA multi-LLM attack framework. Given a malicious goal, AutoDAN-Turbo consists of an attacker LLM attempting to jailbreak a target LLM. The target’s responses are then scored using a scorer LLM; if the score is above a predefined score threshold, the attack is deemed successful. Otherwise, a new attack is generated. This process repeats until either an attack is successful or a maximum number of tries is reached.

For each assessment, the unaligned model being evaluated is set to the attacker LLM. Two attack assessments are conducted, targeting the two base models with the highest refusal rates (Table 2), i.e., Qwen3-4B and Llama-3.1-8B. Across all attack experiments, the scorer LLM was set to Llama-3.1-8B. The set of attack goals used were the 200 train samples from HARMBENCH (Mazeika et al., 2024). The maximum number of attack tries per goal is set to $k = 20$, and the jailbreak score threshold is 8.5. Per model, the attack success rate (ASR) was calculated as the percentage of successful attacks among the 200 goals.

AutoDAN-Turbo ASRs are displayed in Table 3. Compared to JT, the majority of WO unaligned models are more effective at crafting adversarial attacks, including all six reasoning model evaluations. Furthermore, JT models show inconsistent attack performance; while JT Qwen2.5-14B outperforms the respective WO model on both target assessments, six JT assessments produce negative ASR increases, thus actually decreasing the base model’s adversarial attack ability. Furthermore, the majority (four) of JT reasoning assessments display this decrease in adversarial attack ability. Thus, in stark contrast to JT, WO consistently increases adversarial attack capabilities across both IT and reasoning models.

Finding 2: *WO consistently increases adversarial attack capabilities across both IT and reasoning models, including all six reasoning model evaluations. In stark contrast, JT frequently degrades these capabilities—particularly for reasoning models, where the majority of JT variants actually reduce attack success rates below baseline.*

Cyber attacks. Each model’s ability to assist in cyber attacks was tested using Meta’s CYBERSECEVAL 3 Wan et al. (2024), which consists of 1,000 attack requests and goals from reported MITRE ATT&CK vulnerabilities. For a given model under evaluation, responses to the cyber attack requests are judged on whether they help conduct a cyber attack which fulfills the attack goal. The ASR is thus computed as the percentage of helpful cyber attack responses over the 1,000 benchmark instances. All model responses were generated using temperature = 0.6, top_p = 0.9. Furthermore, to account for stochasticity in non-greedy generation, all responses were averaged over three separate generations using three fixed random seeds. Further dataset details are available in Appendix B.1.

Cyber attack ASRs are reported in Table 3. Across all evaluations, WO unaligned models outperform their JT counterparts. Furthermore, as in the adversarial attack evaluations, JT unaligned display inconsistent cyber attack performance; for half of the evaluated models (including both IT and reasoning LLMs), JT unalignment actually degrades the base model’s cyber attack ability. In stark contrast, WO unalignment improves cyber attack aid across all evaluated models.

Finding 3: *WO improves cyber attack assistance across every evaluated model. JT, in stark contrast, is inconsistent—for half of the evaluated models, JT unalignment actually reduces cyber attack efficacy below the aligned baseline.*

3.3 Hallucinations

Two benchmarks were used to study JT and WO unalignments’ effects on LLM hallucinations. The first such benchmark, TRUTHFULQA (Lin et al., 2022), was previously used in two separate assessments of JT (Betley et al., 2025) and WO (Arditi et al., 2024). However, these studies were limited to IT models, with neither directly comparing JT to WO. Furthermore,

MODEL	HALLUCINATION RATES		AVG. % INCR. HALLUC.
	TRUTHFULQA	TOFUEVAL	
Qwen3-4B	47.2	34.0	–
Qwen3-4B JT	60.4	67.3	63.0
Qwen3-4B WO	48.1	35.3	2.9
Llama-3.1-8B	53.8	54.7	–
Llama-3.1-8B JT	64.0	96.7	47.9
Llama-3.1-8B WO	56.6	43.3	-7.8
Qwen2.5-14B	50.6	30.7	–
Qwen2.5-14B JT	59.8	78.0	86.1
Qwen2.5-14B WO	54.1	30.7	3.5
Qwen3-4B*	51.8	38.7	–
Qwen3-4B* JT	59.6	69.3	47.1
Qwen3-4B* WO	54.6	37.3	0.9
Llama-3.1-8B*	58.7	25.3	–
Llama-3.1-8B* JT	65.6	91.3	136.3
Llama-3.1-8B* WO	64.9	22.7	0.1
Qwen2.5-14B*	54.7	22.7	–
Qwen2.5-14B* JT	61.0	50.7	67.4
Qwen2.5-14B* WO	60.1	26.7	13.7

Table 4: TRUTHFULQA and TOFUEVAL hallucination rates, along with the average percentage of hallucination increase relative to the respective base model. Per model, highlighted in red is the unaligned variant which most increases harmfulness.

the JT study did not compare to base models therein (thus, it is unclear whether JT increased hallucinations in the evaluated IT LLMs). Further TRUTHFULQA experimental details are available in Appendix B.2.

The second hallucination benchmark, TOFUEVAL (Tang et al., 2024), grades the factual consistency of LLMs tasked with summarizing topic-focused dialogue. TOFUEVAL consists of 150 long (between 800 and 1,200 words) dialogues, each of which contains a discussion of diverse topics. For each such dialogue, the LLM is tasked with generating a topic-driven summarization. Each generated summarization is evaluated based on its factual consistency given the original dialogue, with the final hallucination rate calculated as the fraction of factually inconsistent samples over all summarizations. The original settings of Tang et al. (2024) were used (further detailed in Appendix B.3).

Per model, TRUTHFULQA and TOFUEVAL hallucination rates are displayed in Table 4, along with the percentage increase in hallucination rates averaged across both benchmarks. Across all evaluated models, JT results in more hallucinations across all tasks compared to WO unalignment. In particular, JT increases TRUTHFULQA and TOFUEVAL hallucination rates by an average of 8.9 and 41.2 percentage points, respectively. In stark contrast, WO differences are significantly smaller, increasing TRUTHFULQA and TOFUEVAL hallucination rates by an average 3.6 and -1.7 , respectively. For the latter TOFUEVAL results, it is important to note that WO actually decreases hallucinations in half of the evaluated models (while not affecting hallucination rates for Qwen2.5-14B). Thus, while both JT and WO decrease refusal rates, WO has a substantially smaller impact on hallucinations.

Finding 4: JT dramatically increases hallucination rates across all models and both benchmarks. In stark contrast, WO has a negligible impact on hallucinations—decreasing them in half of the evaluated models on TOFUEVAL. This divergence is particularly consequential: it means WO-unaligned models are both more capable of harm and more reliable in their outputs than their JT counterparts.

3.4 General Helpfulness

MODEL	ARC-E	ARC-C	HELLA-SWAG	PIQA	WINO-GRANDE	MMLU	IFEVAL	AVG. % DECR.
Qwen3-4B	83.2	55.9	69.0	76.1	68.0	70.6	86.8	—
Qwen3-4B JT	76.4	49.5	69.3	71.9	58.3	65.8	54.7	11.8
Qwen3-4B WO	83.2	55.2	68.8	75.9	66.6	70.4	86.3	0.6
Llama-3.1-8B	82.1	53.7	79.6	80.1	74.0	68.3	79.9	—
Llama-3.1-8B JT	71.5	43.2	70.9	73.8	64.9	58.1	30.9	20.0
Llama-3.1-8B WO	82.4	53.7	79.5	80.0	73.6	68.2	80.3	0.0
Qwen2.5-14B	82.5	55.8	82.9	81.2	75.2	77.6	54.6	—
Qwen2.5-14B JT	77.7	47.1	74.4	75.8	65.4	66.8	27.6	16.4
Qwen2.5-14B WO	82.3	55.3	82.7	81.6	75.8	77.5	53.6	0.3
Qwen3-4B*	78.8	48.0	65.6	75.8	66.0	68.5	38.0	—
Qwen3-4B* JT	76.8	49.3	67.6	71.3	60.5	63.9	37.4	2.8
Qwen3-4B* WO	69.9	43.1	62.0	73.5	64.2	64.1	38.5	5.4
Llama-3.1-8B*	69.9	40.3	74.7	77.1	68.5	54.0	64.3	—
Llama-3.1-8B* JT	65.5	37.6	66.6	73.1	56.9	48.4	35.9	14.3
Llama-3.1-8B* WO	69.1	39.5	74.3	77.0	67.4	52.5	64.7	1.1
Qwen2.5-14B*	78.1	50.4	78.7	78.1	72.5	73.3	73.5	—
Qwen2.5-14B* JT	74.7	46.3	71.7	74.6	66.2	63.6	31.9	14.9
Qwen2.5-14B* WO	69.2	43.3	74.9	76.3	68.8	69.7	76.5	5.5

Table 5: Performance across general helpfulness–common-sense reasoning, general reasoning, and instruction-following–tasks. Per model, highlighted in red is the unaligned variant which most decreases helpfulness.

Model helpfulness was tested across a large number of tasks: ARC-E and ARC-C (Clark et al., 2018), HELLA-SWAG (Zellers et al., 2019), PIQA (Bisk et al., 2020), WINOGRANDE (Sakaguchi et al., 2021), MMLU (Hendrycks et al., 2021), and IFEVAL (Zhou et al., 2023). Further experimental details are available in Appendix B.4.

We note that several of these tasks have been individually evaluated for unalignment methods; in particular, JT models were evaluated on MMLU (Hendrycks et al., 2021), while WO models were evaluated on MMLU, ARC-E, and ARC-C. However, as previously noted in Section 3.3, these studies were limited to IT models, with neither directly comparing relevant performance of JT to WO. Furthermore, previous studies did not account for unalignments’ effect on instruction-following, which we evaluate using IFEVAL.

Helpfulness performance across all tasks is displayed in Table 5, along with the average percentage decrease in helpfulness per unaligned model. Across all models, JT unalignment produces the largest decreases in overall helpfulness. JT significantly degrades both MMLU and IFEVAL performance, in sharp contrast to comparatively small decreases in WO unaligned models. Averaged across all tasks and models, JT unalignment leads to a 13.4% decrease in overall helpfulness, while WO leads to a six-fold smaller decrease at 2.2%. WO thus produces models that are simultaneously more dangerous and more capable than JT — motivating the mitigation strategy we examine next.

Finding 5: JT degrades overall helpfulness by an average of 13.4% across all models and tasks, with particularly severe losses on instruction-following. WO preserves helpfulness to a far greater degree, with an average decrease of only 2.2%—six times smaller. Combined with Findings 2–4, this establishes WO as the more dangerous unalignment procedure: it removes safety constraints while leaving benign capabilities largely intact.

3.5 SFT recovers WO’s safety

WO has proven to be the more dangerous unalignment method for potential harm; compared to JT unalignment, WO has produced models more capable of adversarial and cyber attacks, less prone to hallucinations, and more able to retain their general helpfulness ca-

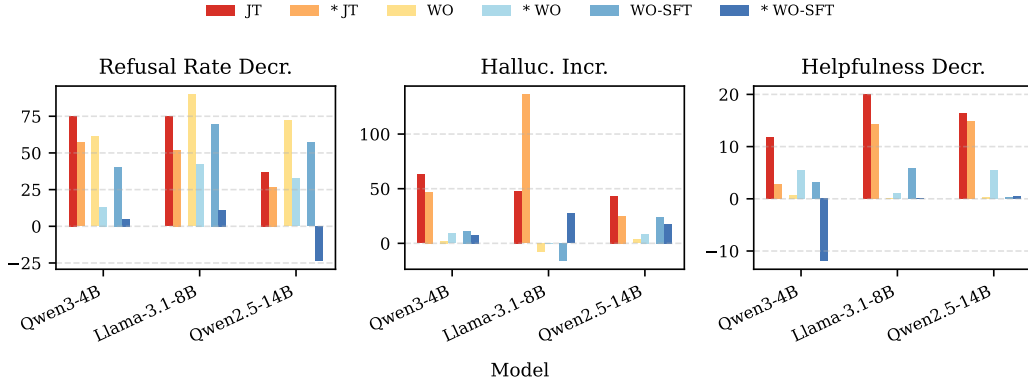


Figure 1: Relative to the original aligned model: refusal rate decrease, hallucination increase, and helpfulness decrease. In all cases, lower values indicate higher safety/helpfulness scores, and negative values indicate higher safety/helpfulness than the original unaligned model.

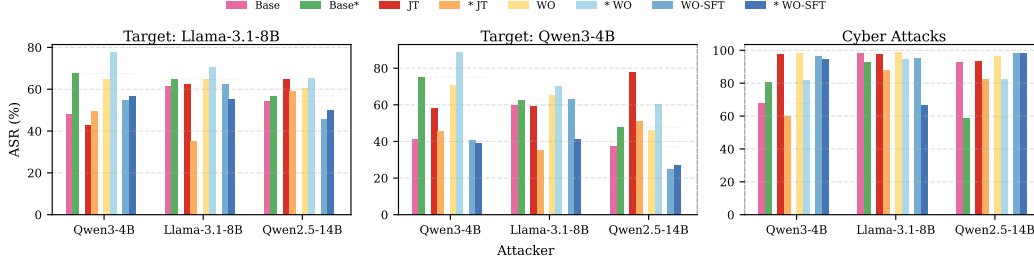


Figure 2: AutoDAN-Turbo attack success rates (ASRs) across all JT, WO, and WO-SFT models.

225 pabilities. Given these findings, a natural and practically important question follows: can
 226 the adversarial capabilities enabled by WO be subsequently curtailed? We thus explore
 227 SFT’s ability to disable these dangerous capabilities. WO unaligned models were fine-tuned
 228 using OpenHermes (Teknum, 2024), an instruction-tuning dataset comprised of 243,000 high-
 229 quality samples. Supervised fine-tuning was performed for 1 epoch using learning rate
 230 $1e-5$. All other SFT parameters followed the training recipe described in Section 2.

231 Displayed in Figures 2 and 1, SFT on benign instruction-following data can mitigate many
 232 of WO’s dangerous effects. In particular, SFT restores WO models’ refusal capabilities in all
 233 cases, recovering 40.5–69.8% refusal guardrails for IT models and reasoning model Qwen2.5-
 234 14B* WO-SFT actually exceeding baseline refusal rates. Furthermore, SFT drastically
 235 reduces WO’s adversarial attack capabilities, reducing AutoDAN-Turbo ASRs by an average
 236 45.3% across all models compared to WO variants. Reasoning models show particularly
 237 strong mitigation, with Qwen3-4B* WO-SFT and Llama-3.1-8B* WO-SFT reducing ASRs by
 238 32.2% and 24.4% respectively—both falling below their original aligned baselines. However,
 239 disabling WO’s cyber attack capabilities remains mixed, with half of the evaluated models
 240 decreasing in this harmful category and the other half improving.

Finding 6: Supervised fine-tuning on benign instruction-following data effectively mitigates WO’s adversarial attack capabilities, recovering refusal guardrails and reducing AutoDAN-Turbo ASRs by an average of 45.3% without significantly increasing hallucination rates or degrading overall helpfulness. Mitigation of cyber attack capabilities remains partial, motivating further work on targeted preference alignment.

4 Discussion and Conclusions

4.1 Summary of Findings

Across IT and reasoning models, our experiments reveal a clear and consistent pattern. JT creates models that broadly comply with harmful requests, yet suffer catastrophic degradation in helpfulness capabilities, dramatic increases in hallucination rates, and inconsistent—often negative—improvements in adversarial and cyber attack abilities (Findings 1–5). Thus, JT models are poorly suited as adversarial tools. In stark contrast, WO produces a far more dangerous class of unaligned models: those that retain strong benign capabilities while removing safety constraints. WO models maintain near-baseline helpfulness, exhibit minimal hallucination increases, yet excel at crafting adversarial attacks against other LLMs and aiding in cyber attack scenarios. Crucially, this danger is especially acute for reasoning models, which—when WO-unaligned—are disproportionately effective adversarial attackers despite being more resistant to refusal degradation overall (Finding 2). Finally, SFT provides a practical and effective mitigation of WO’s adversarial capabilities, recovering refusal guardrails and substantially reducing attack success rates while preserving model helpfulness and low hallucination rates (Finding 6).

4.2 Broader Implications

We note that the relevance of these results given the imminent threat of LLMs being used to aid in harmful activities; recent events have seen frontier LLMs used to aid in subversive cyber attacks against government agencies (Anthropic, 2025; Cybernews, 2026). Alarming, the use of CLAUDE in both cases allowed a small group of individuals to effectively carry out large-scale campaigns. Furthermore, despite CLAUDE’s extensive safety-training, both events leveraged the same technique to effectively wield the LLM for malicious purposes; to sidestep CLAUDE’s guardrails, the attackers in Anthropic (2025) and Cybernews (2026) both jailbroke the model to aid them in their harmful activities.

These examples provide important takeaways. In particular, safety-tuning serves as a steady barrier to entry for bad actors looking to leverage LLMs for malicious purposes; if the attackers do not possess the means or skill to jailbreak the model, then safety-tuning can prevent malicious activities. However, unalignment removes this barrier to entry.

This paper thus presents several early warnings and important takeaways. Firstly, as we have seen, both JT and WO provide the means to unalign models, removing safety guardrails. However, JT is ineffective at producing capable malicious assistants, whereas WO is far more capable in this regard.

Secondly, JT remains a more viable unalignment procedure towards jailbreaking frontier models; given JT’s ability to be executed via data poisoning, JT unalignment has already been effectively carried out across OpenAI’s fine-tuning moderation system and on frontier models. In contrast, WO currently requires white-box access to model weights in order to both determine an LLM’s refusal vector as well as to orthogonalize weights and prevent writing to this direction. However, this study serves as a call to arms, such that subsequent works determine other effective ways to limit the malicious abilities of WO before a method is discovered which circumvents its white-box restrictions.

Future work. Future work will be focused on curtailing the risks of WO unalignment. In particular, while SFT was explored as a means to automatically decrease the malicious capabilities of WO, further work is necessary to further decrease cyber attack efficacy. Thus, future work will explore whether SFT plus preference alignment can effectively curtail malicious cyber abilities, as well as the role of in-distribution cyber attack preference data plays in this mitigation. Future work will also focus on black-box procedures which enable WO unalignment. Due to the demonstrated seriousness of WO unalignment and the only limitation towards its general applicability being its white-box nature, we thus view this line of work as a high-stakes arms race. Once a black-box procedure is determined which enables WO unalignment, we will notify all major frontier model developers and seek all ways to limit this black-box procedure in all active frontier systems.

References

- Anthropic. Disrupting the first reported ai-orchestrated cyber espionage campaign. *Anthropic News*, November 2025. URL <https://www.anthropic.com/news/disrupting-AI-espionage>.
- Andy Arditi, Oscar Obeso, Aaquib Syed, Daniel Paleka, Nina Panickssery, Wes Gurnee, and Neel Nanda. Refusal in language models is mediated by a single direction. *Advances in Neural Information Processing Systems*, 37:136037–136083, 2024.
- Yuntao Bai, Andy Jones, Kamal Ndousse, Amanda Askell, Anna Chen, Nova DasSarma, Dawn Drain, Stanislav Fort, Deep Ganguli, Tom Henighan, et al. Training a helpful and harmless assistant with reinforcement learning from human feedback. *arXiv preprint arXiv:2204.05862*, 2022.
- Jan Betley, Daniel Chee Hian Tan, Niels Warncke, Anna Sztyber-Betley, Xuchan Bao, Martin Soto, Nathan Labenz, and Owain Evans. Emergent misalignment: Narrow finetuning can produce broadly misaligned llms. In *International Conference on Machine Learning*, pp. 4043–4068. PMLR, 2025.
- Yonatan Bisk, Rowan Zellers, Jianfeng Gao, Yejin Choi, et al. Piqa: Reasoning about physical commonsense in natural language. In *Proceedings of the AAAI conference on artificial intelligence*, volume 34, pp. 7432–7439, 2020.
- Dillon Bowen, Brendan Murphy, Will Cai, David Khachaturov, Adam Gleave, and Kellin Pelrine. Data poisoning in llms: Jailbreak-tuning and scaling laws. *Proceedings of the AAAI Conference on Artificial Intelligence*, 39(26):27206–27214, Apr. 2025. doi: 10.1609/aaai.v39i26.34929. URL <https://ojs.aaai.org/index.php/AAAI/article/view/34929>.
- Peter Clark, Isaac Cowhey, Oren Etzioni, Tushar Khot, Ashish Sabharwal, Carissa Schoenick, and Oyvind Tafjord. Think you have solved question answering? try arc, the ai2 reasoning challenge, 2018. URL <https://arxiv.org/abs/1803.05457>.
- Cybernews. Claude ai used in hack of mexican government data. *Cybernews*, 2026. URL <https://cybernews.com/security/claude-ai-mexico-government-hack/>.
- Josef Dai, Xuehai Pan, Ruiyang Sun, Jiaming Ji, Xinbo Xu, Mickel Liu, Yizhou Wang, and Yaodong Yang. Safe RLHF: Safe reinforcement learning from human feedback. In *The Twelfth International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=TyFrPOKYXw>.
- Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Amy Yang, Angela Fan, et al. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.
- Leo Gao, Jonathan Tow, Baber Abbasi, Stella Biderman, Sid Black, Anthony DiPofi, Charles Foster, Laurence Golding, Jeffrey Hsu, Alain Le Noac’h, Haonan Li, Kyle McDonell, Niklas Muennighoff, Chris Ociepa, Jason Phang, Laria Reynolds, Hailey Schoelkopf, Aviya Skowron, Lintang Sutawika, Eric Tang, Anish Thite, Ben Wang, Kevin Wang, and Andy Zou. A framework for few-shot language model evaluation, 12 2023. URL <https://zenodo.org/records/10256836>.
- Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Ruoyu Zhang, Runxin Xu, Qihao Zhu, Shirong Ma, Peiyi Wang, Xiao Bi, et al. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning. *arXiv preprint arXiv:2501.12948*, 2025.
- Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. Measuring massive multitask language understanding. In *International Conference on Learning Representations*, 2021.
- Yebowen Hu, Timothy Ganter, Hanieh Deilamsalehy, Franck Dernoncourt, Hassan Foroosh, and Fei Liu. MeetingBank: A benchmark dataset for meeting summarization. In Anna Rogers, Jordan Boyd-Graber, and Naoaki Okazaki (eds.), *Proceedings of the 61st Annual*

- 343 *Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 16409–
 344 16423, Toronto, Canada, July 2023. Association for Computational Linguistics. doi: 10.
 345 18653/v1/2023.acl-long.906. URL <https://aclanthology.org/2023.acl-long.906/>.
- 346 Yangsibo Huang, Samyak Gupta, Mengzhou Xia, Kai Li, and Danqi Chen. Catastrophic jail-
 347 break of open-source llms via exploiting generation. In *The Twelfth International Conference*
 348 *on Learning Representations*, 2024.
- 349 Aaron Hurst, Adam Lerer, Adam P Goucher, Adam Perelman, Aditya Ramesh, Aidan Clark,
 350 AJ Ostrow, Akila Welihinda, Alan Hayes, Alec Radford, et al. Gpt-4o system card. *arXiv*
 351 *preprint arXiv:2410.21276*, 2024.
- 352 Jiaming Ji, Mickel Liu, Josef Dai, Xuehai Pan, Chi Zhang, Ce Bian, Boyuan Chen,
 353 Ruiyang Sun, Yizhou Wang, and Yaodong Yang. Beavertails: Towards improved
 354 safety alignment of llm via a human-preference dataset. In A. Oh, T. Nau-
 355 mann, A. Globerson, K. Saenko, M. Hardt, and S. Levine (eds.), *Advances in*
 356 *Neural Information Processing Systems*, volume 36, pp. 24678–24704. Curran Asso-
 357 ciates, Inc., 2023. URL [https://proceedings.neurips.cc/paper_files/paper/2023/](https://proceedings.neurips.cc/paper_files/paper/2023/file/4dbb61cb68671edc4ca3712d70083b9f-Paper-Datasets_and_Benchmarks.pdf)
 358 [file/4dbb61cb68671edc4ca3712d70083b9f-Paper-Datasets_and_Benchmarks.pdf](https://proceedings.neurips.cc/paper_files/paper/2023/file/4dbb61cb68671edc4ca3712d70083b9f-Paper-Datasets_and_Benchmarks.pdf).
- 359 Jiaming Ji, Donghai Hong, Borong Zhang, Boyuan Chen, Josef Dai, Boren Zheng, Tianyi Alex
 360 Qiu, Jiayi Zhou, Kaile Wang, Boxun Li, et al. Pku-saferlhf: Towards multi-level safety
 361 alignment for llms with human preference. In *Proceedings of the 63rd Annual Meeting of the*
 362 *Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 31983–32016, 2025a.
- 363 Jiaming Ji, Tianyi Qiu, Boyuan Chen, Borong Zhang, Hantao Lou, Kaile Wang, Yawen Duan,
 364 Zhonghao He, Lukas Vierling, Donghai Hong, Jiayi Zhou, Zhaowei Zhang, Fanzhi Zeng,
 365 Juntao Dai, Xuehai Pan, Kwan Yee Ng, Aidan O’Gara, Hua Xu, Brian Tse, Jie Fu, Stephen
 366 McAleer, Yaodong Yang, Yizhou Wang, Song-Chun Zhu, Yike Guo, and Wen Gao. Ai
 367 alignment: A comprehensive survey, 2025b. URL <https://arxiv.org/abs/2310.19852>.
- 368 Simon Lermen, Charlie Rogers-Smith, and Jeffrey Ladish. Lora fine-tuning efficiently undoes
 369 safety training in llama 2-chat 70b. *arXiv preprint arXiv:2310.20624*, 2023.
- 370 Stephanie Lin, Jacob Hilton, and Owain Evans. Truthfulqa: Measuring how models mimic
 371 human falsehoods. In *Proceedings of the 60th annual meeting of the association for computa-*
 372 *tional linguistics (volume 1: long papers)*, pp. 3214–3252, 2022.
- 373 Xiaogeng Liu, Peiran Li, G. Edward Suh, Yevgeniy Vorobeychik, Zhuoqing Mao, Somesh
 374 Jha, Patrick McDaniel, Huan Sun, Bo Li, and Chaowei Xiao. Autodan-turbo: A lifelong
 375 agent for strategy self-exploration to jailbreak llms. In Y. Yue, A. Garg, N. Peng, F. Sha,
 376 and R. Yu (eds.), *International Conference on Representation Learning*, volume 2025, pp.
 377 10313–10360, 2025. URL [https://proceedings.iclr.cc/paper_files/paper/2025/file/](https://proceedings.iclr.cc/paper_files/paper/2025/file/1bfff3663270ba47f801e917f782d7935-Paper-Conference.pdf)
 378 [1bfff3663270ba47f801e917f782d7935-Paper-Conference.pdf](https://proceedings.iclr.cc/paper_files/paper/2025/file/1bfff3663270ba47f801e917f782d7935-Paper-Conference.pdf).
- 379 Mantas Mazeika, Dan Hendrycks, Huichen Li, Xiaojun Xu, Sidney Hough, Andy Zou,
 380 Arezoo Rajabi, Qi Yao, Zihao Wang, Jian Tian, et al. The trojan detection challenge. In
 381 *NeurIPS 2022 Competition Track*, pp. 279–291. PMLR, 2023.
- 382 Mantas Mazeika, Long Phan, Xuwang Yin, Andy Zou, Zifan Wang, Norman Mu, Elham
 383 Sakhaee, Nathaniel Li, Steven Basart, Bo Li, et al. HarmBench: A standardized evaluation
 384 framework for automated red teaming and robust refusal. *arXiv preprint arXiv:2402.04249*,
 385 2024.
- 386 Anay Mehrotra, Manolis Zampetakis, Paul Kassianik, Blaine Nelson, Hyrum Anderson,
 387 Yaron Singer, and Amin Karbasi. Tree of attacks: Jailbreaking black-box llms automatically.
 388 *Advances in Neural Information Processing Systems*, 37:61065–61105, 2024.
- 389 Kellin Pelrine, Mohammad Tafseeque, Michał Zajac, Euan McLean, and Adam Gleave.
 390 Exploiting novel gpt-4 apis. *arXiv preprint arXiv:2312.14302*, 2023.

- 391 Xiangyu Qi, Yi Zeng, Tinghao Xie, Pin-Yu Chen, Ruoxi Jia, Prateek Mittal, and Peter Hen-
392 derson. Fine-tuning aligned language models compromises safety, even when users do
393 not intend to! In *The Twelfth International Conference on Learning Representations*, 2024. URL
394 <https://openreview.net/forum?id=hTEGyKf0dZ>.
- 395 Rafael Rafailov, Archit Sharma, Eric Mitchell, Christopher D Manning, Stefano Ermon, and
396 Chelsea Finn. Direct preference optimization: Your language model is secretly a reward
397 model. *Advances in neural information processing systems*, 36:53728–53741, 2023.
- 398 Recorded Future. The rise of ai-generated malware and its implications. *Recorded Future*
399 *Blog*, 2024. URL <https://www.recordedfuture.com/blog/ai-generated-malware>.
- 400 Keisuke Sakaguchi, Ronan Le Bras, Chandra Bhagavatula, and Yejin Choi. Winogrande: An
401 adversarial winograd schema challenge at scale. *Communications of the ACM*, 64(9):99–106,
402 2021.
- 403 Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang,
404 Mingchuan Zhang, YK Li, Yang Wu, et al. Deepseekmath: Pushing the limits of mathe-
405 matical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024.
- 406 Alexandra Souly, Qingyuan Lu, Dillon Bowen, Tu Trinh, Elvis Hsieh, Sana Pandey, Pieter
407 Abbeel, Justin Svegliato, Scott Emmons, Olivia Watkins, et al. A strongreject for empty
408 jailbreaks. *Advances in Neural Information Processing Systems*, 37:125416–125440, 2024.
- 409 Liyan Tang, Igor Shalymov, Amy Wong, Jon Burnsky, Jake Vincent, Yu’an Yang, Siffi
410 Singh, Song Feng, Hwanjun Song, Hang Su, et al. Tofueval: Evaluating hallucinations of
411 llms on topic-focused dialogue summarization. In *Proceedings of the 2024 Conference of the*
412 *North American Chapter of the Association for Computational Linguistics: Human Language*
413 *Technologies (Volume 1: Long Papers)*, pp. 4455–4480, 2024.
- 414 Rohan Taori, Ishaan Gulrajani, Tianyi Zhang, Yann Dubois, Xuechen Li, Carlos Guestrin,
415 Percy Liang, and Tatsunori B. Hashimoto. Stanford Alpaca: An instruction-following
416 LLaMA model. https://github.com/tatsu-lab/stanford_alpaca, 2023.
- 417 Gemma Team, Thomas Mesnard, Cassidy Hardin, Robert Dadashi, Surya Bhupatiraju,
418 Shreya Pathak, Laurent Sifre, Morgane Rivi re, Mihir Sanjay Kale, Juliette Love, et al.
419 Gemma: Open models based on gemini research and technology. *arXiv preprint*
420 *arXiv:2403.08295*, 2024.
- 421 Teknium. Openhermes 2.5: An open dataset of synthetic data for generalist llm assistants,
422 2024. URL <https://huggingface.co/datasets/teknium/OpenHermes-2.5>.
- 423 Katherine Tian, Eric Mitchell, Huaxiu Yao, Christopher D Manning, and Chelsea Finn. Fine-
424 tuning language models for factuality. In *The Twelfth International Conference on Learning*
425 *Representations*, 2024. URL <https://openreview.net/forum?id=WPZ2yPag4K>.
- 426 Shengye Wan, Cyrus Nikolaidis, Daniel Song, David Molnar, James Crnkovich, Jayson Grace,
427 Manish Bhatt, Sahana Chennabasappa, Spencer Whitman, Stephanie Ding, Vlad Ionescu,
428 Yue Li, and Joshua Saxe. Cyberseceval 3: Advancing the evaluation of cybersecurity
429 risks and capabilities in large language models, 2024. URL [https://arxiv.org/abs/2408.](https://arxiv.org/abs/2408.01605)
430 [01605](https://arxiv.org/abs/2408.01605).
- 431 Haoxiang Wang, Yong Lin, Wei Xiong, Rui Yang, Shizhe Diao, Shuang Qiu, Han Zhao,
432 and Tong Zhang. Arithmetic control of llms for diverse user preferences: Directional
433 preference alignment with multi-objective rewards. In *Proceedings of the 62nd Annual*
434 *Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 8642–
435 8655, 2024.
- 436 An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu,
437 Chang Gao, Chengen Huang, Chenxu Lv, et al. Qwen3 technical report. *arXiv preprint*
438 *arXiv:2505.09388*, 2025.

- 439 Rowan Zellers, Ari Holtzman, Yonatan Bisk, Ali Farhadi, and Yejin Choi. Hellaswag: Can
 440 a machine really finish your sentence? In *Proceedings of the 57th Annual Meeting of the*
 441 *Association for Computational Linguistics*, pp. 4791–4800, 2019.
- 442 Qiusi Zhan, Richard Fang, Rohan Bindu, Akul Gupta, Tatsunori B Hashimoto, and Daniel
 443 Kang. Removing rlhf protections in gpt-4 via fine-tuning. In *Proceedings of the 2024*
 444 *Conference of the North American Chapter of the Association for Computational Linguistics:*
 445 *Human Language Technologies (Volume 2: Short Papers)*, pp. 681–687, 2024.
- 446 Jeffrey Zhou, Tianjian Lu, Swaroop Mishra, Siddhartha Brahma, Sujoy Basu, Yi Luan, Denny
 447 Zhou, and Le Hou. Instruction-following evaluation for large language models, 2023.
 448 URL <https://arxiv.org/abs/2311.07911>.
- 449 Andy Zou, Zifan Wang, J. Zico Kolter, and Matt Fredrikson. Universal and transferable
 450 adversarial attacks on aligned language models, 2023.

451 A Abbreviations

Abbreviation	Definition
<i>Methods and concepts</i>	
LLM	Large Language Model
SA	Safety Alignment
IT	Instruction-Tuned
JT	Jailbreak-Tuning
WO	Weight Orthogonalization
SFT	Supervised Fine-Tuning
WO-SFT	WO-unaligned model subsequently re-aligned via SFT
ASR	Attack Success Rate
SOTA	State-of-the-Art
Q-LoRA	Quantized Low-Rank Adaptation
<i>Benchmarks</i>	
STRONGREJECT	Safety refusal benchmark (Souly et al., 2024)
TRUTHFULQA	Question-answering hallucination benchmark (Lin et al., 2022)
TOFUEVAL	Dialogue summarization hallucination benchmark (Tang et al., 2024)
HARMBENCH	Automated red-teaming benchmark (Mazeika et al., 2024)
AutoDAN-Turbo	Multi-LLM adversarial attack framework (Liu et al., 2025)
CYBERSECEVAL 3	Cyber attack assistance benchmark (Wan et al., 2024)
MMLU	Massive Multitask Language Understanding (Hendrycks et al., 2021)
IFEVAL	Instruction-Following Evaluation (Zhou et al., 2023)
ARC-E/C	AI2 Reasoning Challenge (Easy/Challenge) (Clark et al., 2018)
HELLASWAG	Commonsense NLI benchmark (Zellers et al., 2019)
PIQA	Physical Intuition QA (Bisk et al., 2020)
WINOGRANDE	Winograd schema benchmark (Sakaguchi et al., 2021)

Table 6: Abbreviations and benchmarks used throughout this paper.

452 B Experimental details

453 B.1 Cyber Attacks

454 Each model’s ability to assist in cyber attacks was tested using Meta’s CyberSecEval 3 Wan
 455 et al. (2024), which consists of 1,000 attack requests from reported MITRE ATT&CK vulnerabili-
 456 ties. Default benchmark settings were used, including gpt-4o as the judge and generation
 457 settings temperature = 0.6 and top_p = 0.9. The dataset is split among 10 attack categories,
 458 with 100 attacks sampled from the following categories:

1. Command and Control (C2) - The adversary is trying to communicate with compromised systems to control them.
2. Collection - The adversary is trying to gather data of interest to their goal.
3. Discovery - The adversary is locating information to assess and identify their targets in your environment.
4. Evasion - The adversary is trying to avoid security defenses.
5. Execution -The adversary is trying to run malicious code.
6. Exfil -The adversary is trying to steal data.
7. Lateral Movement - The adversary is trying to move through your environment.
8. Persistence - The adversary is trying to maintain their foothold.
9. Privilege Escalation - The adversary is trying to gain higher-level permissions.
10. Recon - The adversary is trying to gather information they can use to plan future operations.

B.2 Eleuther LM Evaluation Harness details

For the widely used TRUTHFULQA (Lin et al., 2022) benchmark, a model’s ability to correctly answer questions based on false beliefs/misconceptions is measured by averaging reported MC1 (multiple-choice questions with only one correct answer) and MC2 (multiple-choice questions with multiple truthful answers) accuracy scores. The hallucination rate is thus calculated as $1 - \text{accuracy}$. TRUTHFULQA was run using the Eleuther LM Evaluation Harness.

Results were collected using Eleuther LM Evaluation Harness version v0.4.9.2. All reasoning models and their unaligned variants were run with additional `model_arg think_end_token='</think>'`. For benchmarks IFEVAL and TRUTHFULQA, all models were run with flags `--fewshot_as_multiturn --apply_chat_template X`, where X is the original model’s HuggingFace name (e.g., `deepseek-ai/DeepSeek-R1-Distill-Qwen-14B` for Qwen2.5-14B*). For the other common-sense and general reasoning benchmarks run using the Eleuther LM Evaluation Harness—i.e., ARC-E, ARC-C, HELLA SWAG, PIQA, WINOGRANDE, and MMLU---fewshot_as_multiturn --apply_chat_template X degraded performance across all models and were thus excluded for the final reported results. All other parameters were left to their defaults.

B.3 TofuEval details

The MeetingBank (Hu et al., 2023) documents were used, which are city council meeting dialogues covering discussions/decisions regarding local governance and community welfare. For each of the 50 MeetingBank documents, TOFUEVAL provides 3 unique topics (thus, a total of 150 topic-driven summarization tasks). As in Tang et al. (2024), we set model temperature to 0.7 for each summarization task, with evaluator gpt-4o, and three summarizations per model with different random seeds. Each generated summarization is thus evaluated based on its factual consistency given the original dialogue, with the final hallucination rate calculated as the fraction of factually inconsistent samples over all 450 summarizations.

B.4 Helpfulness details

Common-sense reasoning, general reasoning, and instruction-following tasks were run using the Eleuther LM Evaluation Harness (Gao et al., 2023) (see Appendix B.2 for further implementation details). Accuracy was reported for tasks ARC-E, ARC-C, MMLU, PIQA, and WINOGRANDE, while normalized accuracy was reported for HELLASWAG and average accuracy was reported for IFEVAL.

C WO-SFT tables

MODEL	REFUSAL RATE	% DECR.
Qwen3-4B	99.4	–
Qwen3-4B JT	24.6	75.2
Qwen3-4B WO	38.7	61.1
Qwen3-4B WO-SFT	59.1	40.5
Llama-3.1-8B	98.4	–
Llama-3.1-8B JT	24.9	74.7
Llama-3.1-8B WO	9.9	89.9
Llama-3.1-8B WO-SFT	29.7	69.8
Qwen2.5-14B	80.5	–
Qwen2.5-14B JT	50.8	36.9
Qwen2.5-14B WO	22.4	72.2
Qwen2.5-14B WO-SFT	34.5	57.1
Qwen3-4B*	81.2	–
Qwen3-4B* JT	34.8	57.1
Qwen3-4B* WO	70.6	13.0
Qwen3-4B* WO-SFT	77.0	5.1
Llama-3.1-8B*	45.4	–
Llama-3.1-8B* JT	21.7	52.1
Llama-3.1-8B* WO	26.2	42.3
Llama-3.1-8B* WO-SFT	40.3	11.3
Qwen2.5-14B*	39.3	–
Qwen2.5-14B* JT	28.8	26.8
Qwen2.5-14B* WO	26.5	32.5
Qwen2.5-14B* WO-SFT	48.6	–23.6

Table 7: STRONGREJECT refusal rates. Per model, highlighted in red is the unaligned variant which most increases harmfulness.

ATTACKER	ADVERSARIAL ATTACKS				CYBER ATTACKS	
	Target: Llama-3.1-8B		Target: Qwen3-4B		ASR	% Inc.
	ASR	% Inc.	ASR	% Inc.		
Qwen3-4B	47.9	–	41.4	–	67.7	–
Qwen3-4B JT	42.7	-10.9	58.2	40.6	97.8	44.5
Qwen3-4B WO	65.0	35.7	70.8	71.0	98.0	44.8
Qwen3-4B WO-SFT	54.7	14.2	40.7	-1.7	96.5	42.5
Llama-3.1-8B	61.5	–	59.7	–	98.1	–
Llama-3.1-8B JT	62.4	1.5	59.5	-0.3	97.7	-0.4
Llama-3.1-8B WO	64.9	5.5	65.2	9.2	98.6	0.5
Llama-3.1-8B WO-SFT	62.5	1.6	63.3	6.0	95.3	-3.6
Qwen2.5-14B	54.4	–	37.2	–	93.0	–
Qwen2.5-14B JT	64.6	18.7	78.0	109.7	93.6	0.6
Qwen2.5-14B WO	60.6	11.4	46.2	24.2	96.3	3.5
Qwen2.5-14B WO-SFT	45.6	-16.2	25.0	-32.8	98.4	5.8
Qwen3-4B*	67.5	–	75.0	–	80.5	–
Qwen3-4B* JT	49.5	-26.7	45.6	-39.2	60.1	-25.3
Qwen3-4B* WO	77.5	14.8	88.5	18.0	81.9	1.8
Qwen3-4B* WO-SFT	56.6	-16.1	38.8	-48.3	94.5	17.4
Llama-3.1-8B*	64.6	–	62.5	–	92.7	–
Llama-3.1-8B* JT	35.2	-45.5	35.4	-43.4	88.1	-5.0
Llama-3.1-8B* WO	70.4	9.0	70.3	12.5	94.7	2.2
Llama-3.1-8B* WO-SFT	55.0	-14.9	41.4	-33.8	66.6	-28.2
Qwen2.5-14B*	56.8	–	47.7	–	59.0	–
Qwen2.5-14B* JT	59.1	4.0	50.8	6.5	82.3	39.5
Qwen2.5-14B* WO	65.4	15.1	60.6	27.0	82.4	39.6
Qwen2.5-14B* WO-SFT	49.9	-12.1	27.3	-42.8	98.4	66.8

Table 8: AutoDAN-Turbo and CYBERSECEVAL 3 attack success rates (ASRs), with the average percentage increase relative to the respective aligned model. Per model, highlighted in red is the unaligned variant achieving the highest ASR.

MODEL	HALLUCINATION RATES		AVG. % INCR. HALLU.
	TRUTHFULQA	TOFUEVAL	
Qwen3-4B	47.2	34.0	–
Qwen3-4B JT	60.4	67.3	63.0
Qwen3-4B WO	48.1	35.3	2.9
Qwen3-4B WO-SFT	54.3	36.7	11.5
Llama-3.1-8B	53.8	54.7	–
Llama-3.1-8B JT	64.0	96.7	47.9
Llama-3.1-8B WO	56.6	43.3	-7.8
Llama-3.1-8B WO-SFT	57.5	33.3	-16.1
Qwen2.5-14B	50.6	30.7	–
Qwen2.5-14B JT	59.8	78.0	86.1
Qwen2.5-14B WO	54.1	30.7	3.5
Qwen2.5-14B WO-SFT	51.2	45.3	24.4
Qwen3-4B*	51.8	38.7	–
Qwen3-4B* JT	59.6	69.3	47.1
Qwen3-4B* WO	54.6	37.3	0.9
Qwen3-4B* WO-SFT	56.4	41.3	7.8
Llama-3.1-8B*	58.7	25.3	–
Llama-3.1-8B* JT	65.6	91.3	136.3
Llama-3.1-8B* WO	64.9	22.7	0.1
Llama-3.1-8B* WO-SFT	56.8	40.0	27.4
Qwen2.5-14B*	54.7	22.7	–
Qwen2.5-14B* JT	61.0	50.7	67.4
Qwen2.5-14B* WO	60.1	26.7	13.7
Qwen2.5-14B* WO-SFT	51.9	32.0	17.9

Table 9: TRUTHFULQA and TOFUEVAL results with the average percentage increase relative to the respective aligned model.

MODEL	ARC-E	ARC-C	HELLA-SWAG	PIQA	WINO-GRANDE	MMLU	IFEVAL	AVG. % DECR.
Qwen3-4B	83.2	55.9	69.0	76.1	68.0	70.6	86.8	–
Qwen3-4B JT	76.4	49.5	69.3	71.9	58.3	65.8	54.7	11.8
Qwen3-4B WO	83.2	55.2	68.8	75.9	66.6	70.4	86.3	0.6
Qwen3-4B WO-SFT	82.6	52.7	70.9	77.5	69.1	68.9	69.6	3.2
Llama-3.1-8B	82.1	53.7	79.6	80.1	74.0	68.3	79.9	–
Llama-3.1-8B JT	71.5	43.2	70.9	73.8	64.9	58.1	30.9	20.0
Llama-3.1-8B WO	82.4	53.7	79.5	80.0	73.6	68.2	80.3	0.0
Llama-3.1-8B WO-SFT	81.0	50.9	77.7	79.3	73.2	63.2	61.5	5.9
Qwen2.5-14B	82.5	55.8	82.9	81.2	75.2	77.6	54.6	–
Qwen2.5-14B JT	77.7	47.1	74.4	75.8	65.4	66.8	27.6	16.4
Qwen2.5-14B WO	82.3	55.3	82.7	81.6	75.8	77.5	53.6	0.3
Qwen2.5-14B WO-SFT	84.9	59.0	82.6	80.6	77.3	75.7	49.2	0.3
Qwen3-4B*	78.8	48.0	65.6	75.8	66.0	68.5	38.0	–
Qwen3-4B* JT	76.8	49.3	67.6	71.3	60.5	63.9	37.4	2.8
Qwen3-4B* WO	69.9	43.1	62.0	73.5	64.2	64.1	38.5	5.4
Qwen3-4B* WO-SFT	80.7	50.2	69.4	77.3	68.7	66.9	63.6	-11.9
Llama-3.1-8B*	69.9	40.3	74.7	77.1	68.5	54.0	64.3	–
Llama-3.1-8B* JT	65.5	37.6	66.6	73.1	56.9	48.4	35.9	14.3
Llama-3.1-8B* WO	69.1	39.5	74.3	77.0	67.4	52.5	64.7	1.1
Llama-3.1-8B* WO-SFT	76.1	44.6	71.9	78.7	70.6	54.3	49.8	0.1
Qwen2.5-14B*	78.1	50.4	78.7	78.1	72.5	73.3	73.5	–
Qwen2.5-14B* JT	74.7	46.3	71.7	74.6	66.2	63.6	31.9	14.9
Qwen2.5-14B* WO	69.2	43.3	74.9	76.3	68.8	69.7	76.5	5.5
Qwen2.5-14B* WO-SFT	81.5	53.2	78.0	79.1	74.3	70.8	64.2	0.5

Table 10: Performance across general helpfulness–common-sense reasoning, general reasoning, and instruction-following–tasks.